

S&DS 2300 Heart Disease Prediction Project

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Introduction

Cardiovascular disease is the leading cause of death globally, making early identification of patients at risk a major clinical priority. This report analyzes a dataset of 918 patients across five international medical institutions. Clinical measurements are analyzed to identify predictors of heart disease. In descriptive analysis, regression modeling, and binary logistic regression, we examine relationships between patient characteristics and maximum heart rate and heart disease presence.

Data

The following variables are drawn from the Heart Failure Prediction Dataset sourced from Kaggle (fedesoriano, 2021), containing 746 patients after cleaning.

Variable	Type	Description
Age	Continuous	Patient age (years)
Sex	Categorical	Patient sex (M = Male, F = Female)
ChestPainType	Categorical	Chest pain type (ASY, ATA, NAP, TA)
RestingBP	Continuous	Resting blood pressure (mm Hg)
Cholesterol	Continuous	Serum cholesterol (mm/dl)
FastingBS	Categorical	Fasting blood sugar > 120 mg/dl (1 = Yes, 0 = No)

Variable	Type	Description
RestingECG	Categorical	Resting ECG results (Normal, ST, LVH)
MaxHR	Continuous	Maximum heart rate achieved (bpm)
ExerciseAngina	Categorical	Exercise-induced angina (Y/N)
Oldpeak	Continuous	ST depression induced by exercise
ST_Slope	Categorical	Slope of peak exercise ST segment (Up, Flat, Down)
HeartDisease	Categorical	Heart disease present (1 = Yes, 0 = No)

Data Inspection and Cleaning

All categorical variables are stored as `chr` and were thus converted to factors for use in subsequent analysis.

```
##           Age           Sex ChestPainType      RestingBP
Cholesterol
##           0             0             0             1
172
##      FastingBS      RestingECG      MaxHR ExerciseAngina
Oldpeak
##           0             0             0             0
0
##      ST_Slope      HeartDisease
##           0             0
```

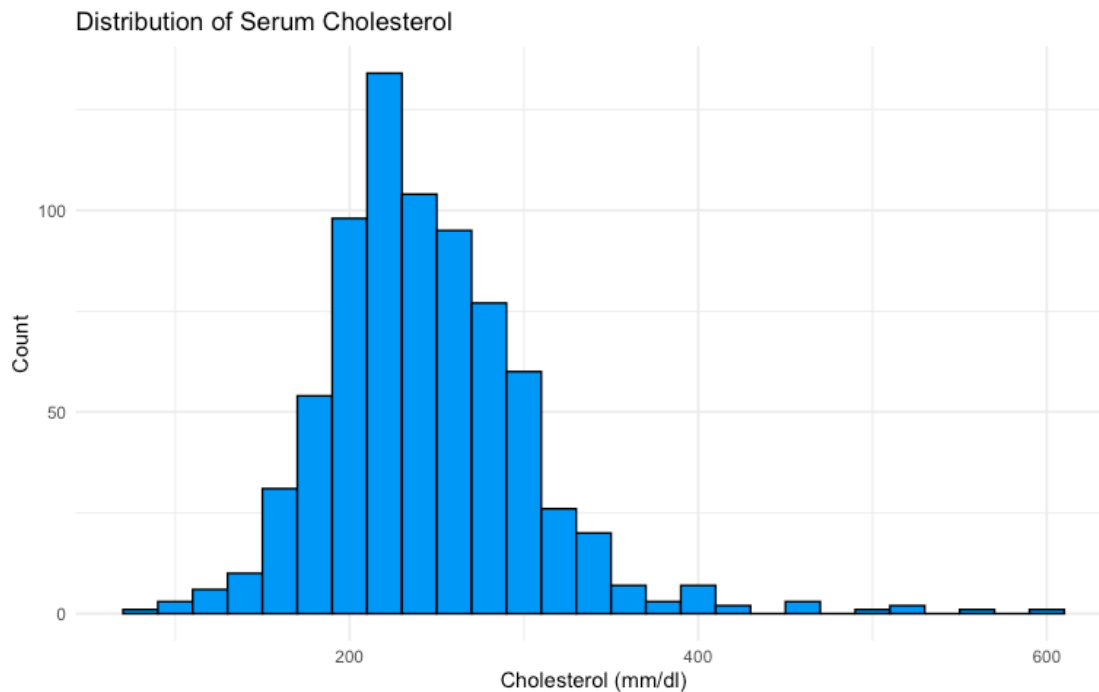
There were 172 cholesterol values and 1 `RestingBP` value of 0. These values are physiologically impossible and were thus replaced with `NA`.

After removing rows with missing values, the final working dataset `heart_clean` contains 746 observations, down from the original 918. You might notice that only 172 rows were removed and not 173 (172 in `Cholesterol` and 1 in `RestingBP`). This is

because one patient had missing values in both variables, and `na.omit()` removes the entire row regardless of how many `NA`'s it contains, so 173 values were replaced with `NA`, which ended up being 172 rows. The dataset required no text manipulation, as variable names and category labels were already consistently formatted upon import.

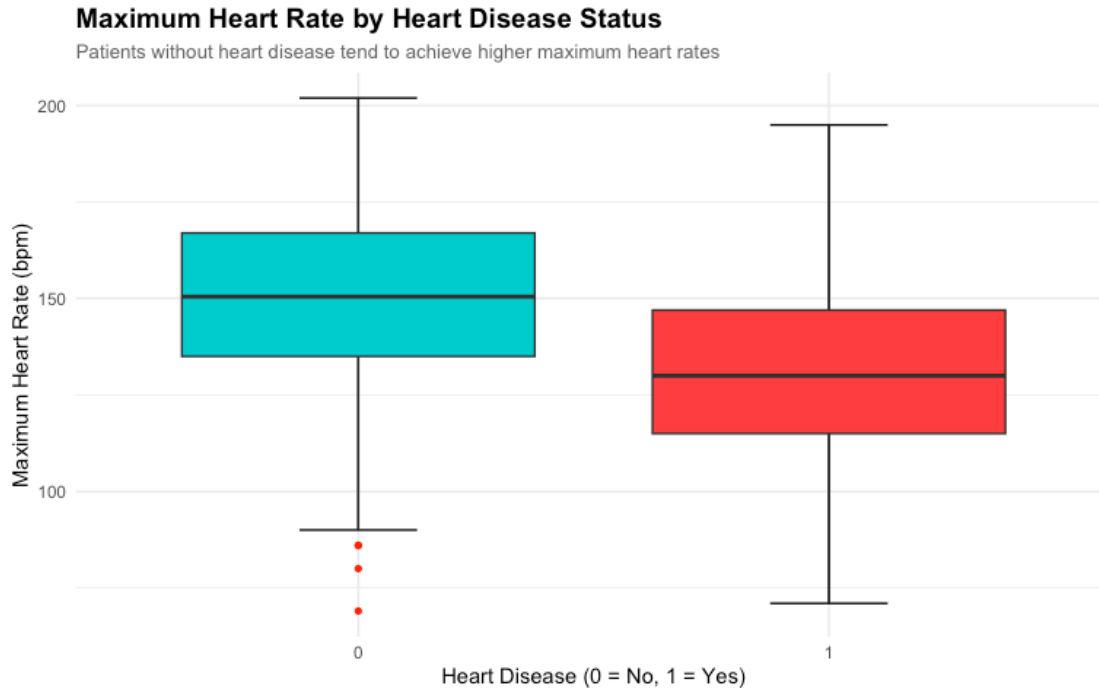
Descriptive Plots

Histogram



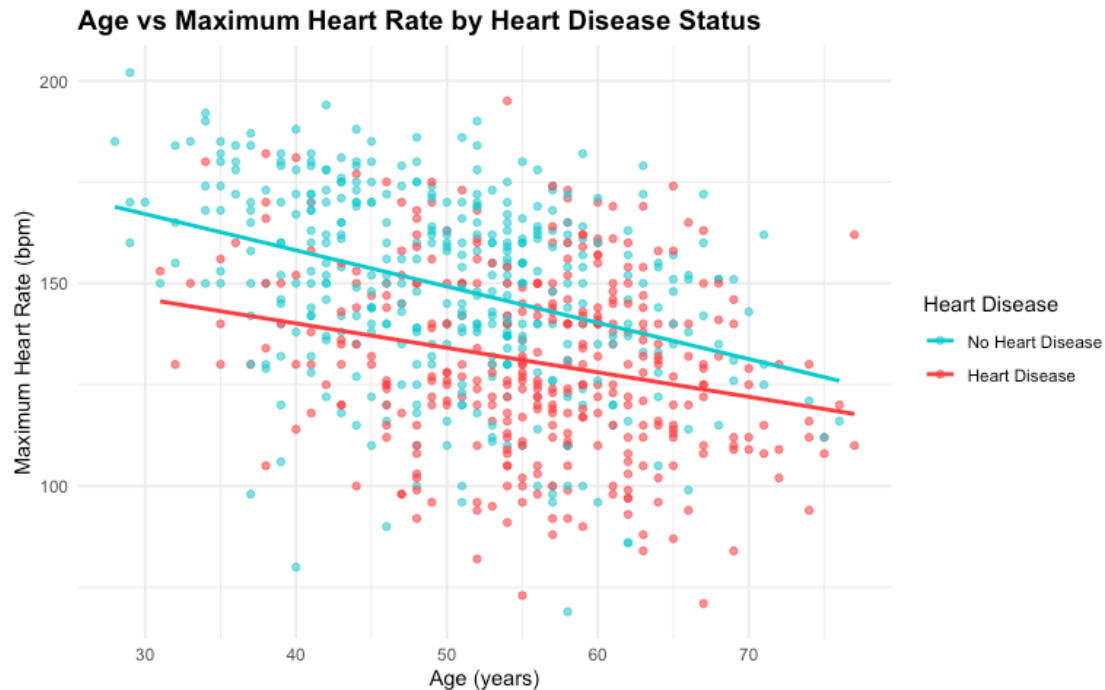
The distribution of serum cholesterol is right-skewed, most patients fall between 150 and 350 mm/dl. A small number of extreme values above 500 mm/dl pull the tail to the right. This skew is worth keeping in mind for later regression analysis, where a log transformation of cholesterol might be appropriate.

Box Plots



Patients without heart disease (group 0) tend to have higher maximum heart rates, with a median of approximately 150 bpm, compared to a median of approximately 130 bpm for patients with heart disease (group 1). This suggests maximum heart rate could be a useful predictor of heart disease, which we'll explore further in the regression analysis.

Scatter Plot



Maximum heart rate shows a clear negative relationship with age for both groups: As age increases, maximum heart rate tends to decrease. Patients without heart disease (blue) generally achieve higher maximum heart rates across all ages compared to those with heart disease (red), with the gap being more pronounced in younger patients. Both regression lines have similar slopes, suggesting the effect of age on maximum heart rate is likely comparable regardless of heart disease status.

Basic Tests

Correlation Test

```
##  
## Pearson's product-moment correlation  
##  
## data: heart_clean$Age and heart_clean$MaxHR  
## t = -11.278, df = 744, p-value < 2.2e-16  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:
```

```
## -0.4417755 -0.3190836
## sample estimates:
##      cor
## -0.3821121
```

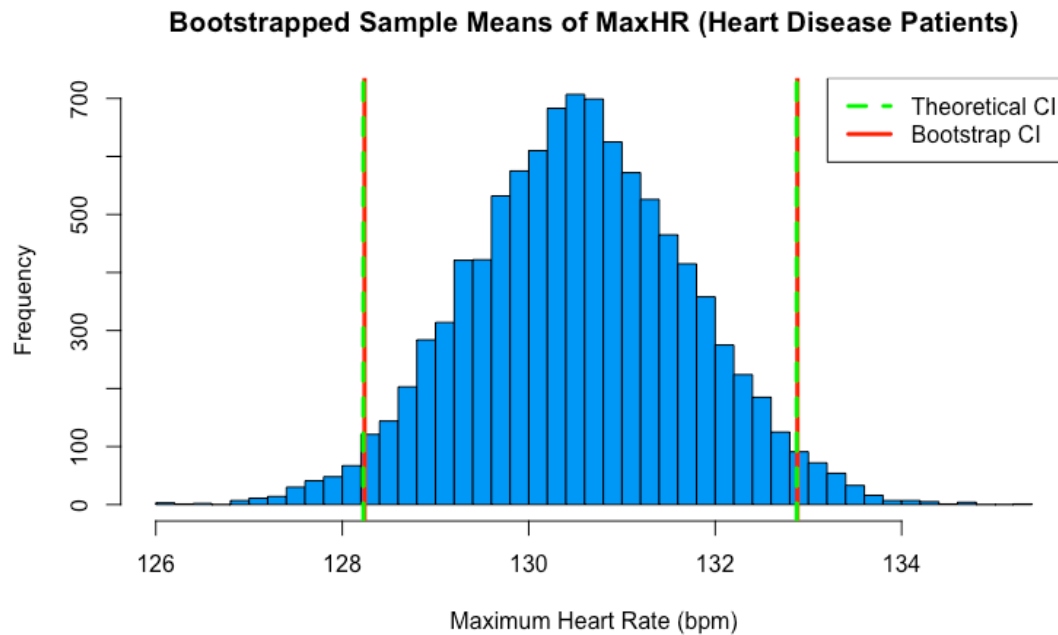
There is a statistically significant, moderate negative correlation between age and maximum heart rate ($r = -0.38$, $p < 0.001$), which indicates that as age increases, maximum heart rate tends to decrease. This is consistent with the pattern we observed in the scatterplot above.

T-Test

```
##
## Welch Two Sample t-test
##
## data: Cholesterol by HeartDisease
## t = -2.833, df = 712.54, p-value = 0.004742
## alternative hypothesis: true difference in means between group 0
and group 1 is not equal to 0
## 95 percent confidence interval:
## -20.811482 -3.773652
## sample estimates:
## mean in group 0 mean in group 1
##      238.7692      251.0618
```

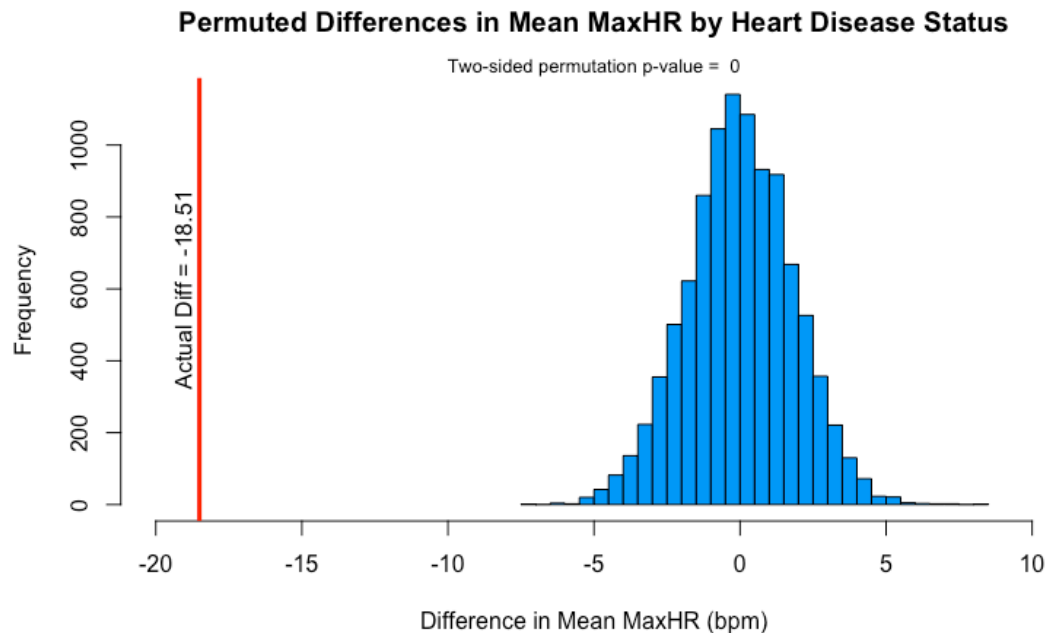
The Welch two-sample t-test identified the difference in mean cholesterol between patients with and without heart disease as statistically significant ($t = -2.83$, $df = 712.54$, $p = 0.005$). The mean was higher in the patients with heart disease (251.1 mm/dl) than in those without (238.8 mm/dl).

Bootstrap



The bootstrapped 95% CI for mean maximum heart rate among heart disease patients was (128.239, 132.882) bpm, which is nearly identical to the theoretical t-test CI (128.226, 132.875) bpm. Given the large sample size, this is expected because with a sample this large, the Central Limit Theorem ensures the sampling distribution of the mean is approximately normal, so the two methods converge.

Permutation Test



A two-sided permutation test was conducted to assess whether the observed difference in mean maximum heart rate between heart disease groups (-18.51 bpm) could have occurred by chance. Across the 10,000 permutations, none produced a difference nearly as extreme as the observed value, yielding a permutation p-value < 0.0001, strong evidence that the difference in mean maximum heart rate between patients with and without heart disease is statistically significant.

Multiple Regression

Description of plan

To predict maximum heart rate, a multiple regression model will be fit using all available predictors as candidate variables: Age, Sex, ChestPainType, RestingBP, Cholesterol, FastingBS, RestingECG, ExerciseAngina, Oldpeak, ST_Slope, and HeartDisease. Manual backwards stepwise regression will be used, the least significant predictor will be removed one at a time until all remaining predictors are significant. For non-significant categorical variables with more than two levels, an

ANOVA F-test will be used to test whether the variable as a whole should or should not be removed.

Backwards Stepwise Regression

Starting from the full model, predictors were removed one at a time in the following order by highest p-value: *RestingBP* ($p = 0.975$), *HeartDisease* ($p = 0.717$), *FastingBS* ($p = 0.590$), *Cholesterol* ($p = 0.397$), and *oldpeak* ($p = 0.310$). For *ST_Slope*, an ANOVA F-test confirmed it should be retained as a whole ($p < 0.001$) despite its individual level p-values being non-significant, and this is further supported by a lower AIC in the model with *ST_Slope* (AIC = 6617.43) compared to the model without it (AIC = 6634.91). The final model summary below.

Final Model

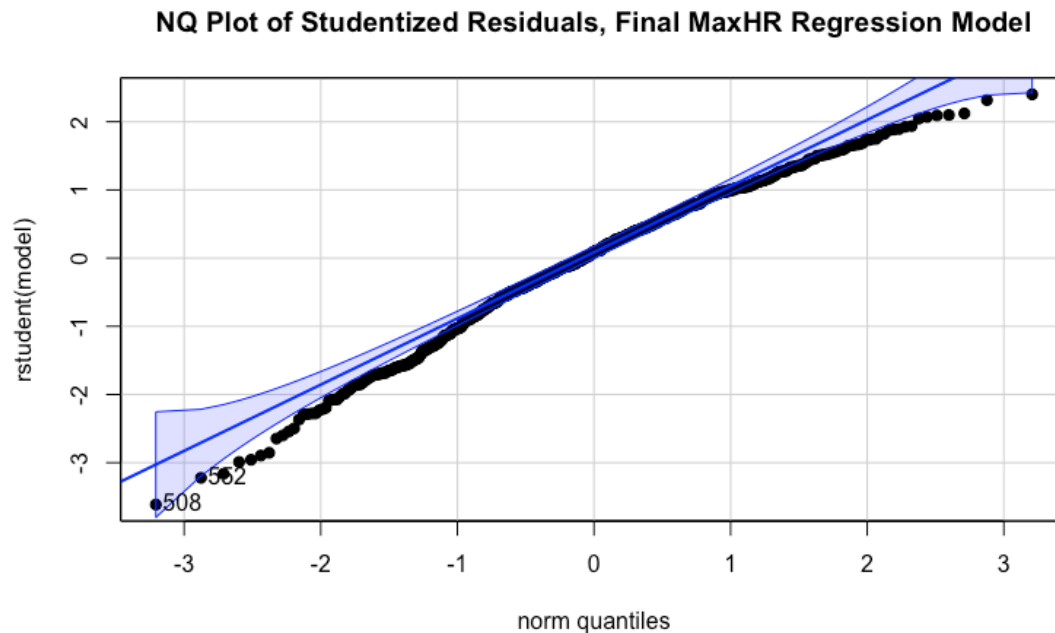
```
##
## Call:
## lm(formula = MaxHR ~ Age + Sex + ChestPainType + RestingECG +
##      ExerciseAngina + ST_Slope, data = heart_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -71.885 -11.472   1.849  14.902  48.136
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   188.17593     6.32928   29.731 < 2e-16 ***
## Age           -0.76645     0.08466   -9.053 < 2e-16 ***
## SexM          -3.56680     1.77435   -2.010 0.044774 *
## ChestPainTypeATA  8.15901     2.21566    3.682 0.000248 ***
## ChestPainTypeNAP  8.49819     2.03721    4.171 3.39e-05 ***
## ChestPainTypeTA  12.69513     3.44018    3.690 0.000241 ***
## RestingECGNormal -8.53309     1.86161   -4.584 5.37e-06 ***
## RestingECGST     -11.52303     2.39248   -4.816 1.78e-06 ***
## ExerciseAnginaY  -6.79777     1.92370   -3.534 0.000435 ***
## ST_SlopeFlat    -3.69474     3.29335   -1.122 0.262281
## ST_SlopeUp       4.76633     3.49972    1.362 0.173641
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.24 on 735 degrees of freedom
```

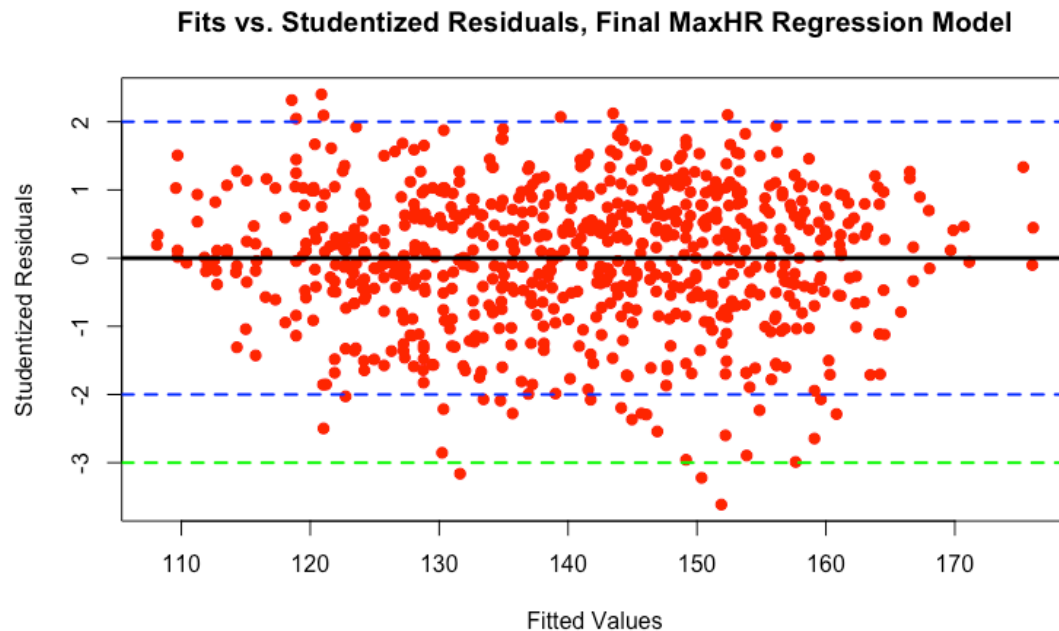
```
## Multiple R-squared:  0.328, Adjusted R-squared:  0.3188  
## F-statistic: 35.87 on 10 and 735 DF,  p-value: < 2.2e-16
```

Results & Discussion

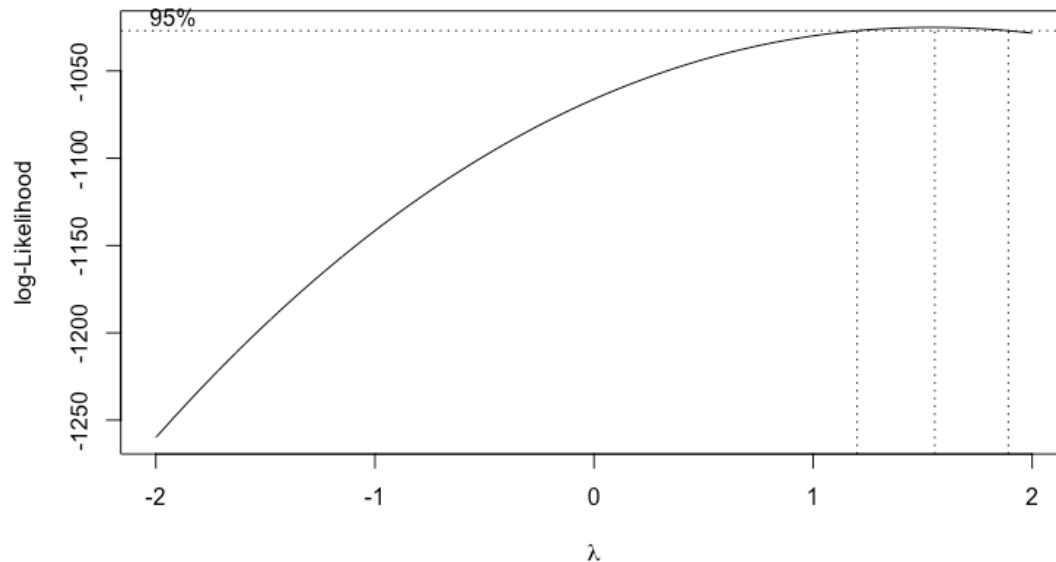
Five predictors were removed during backwards stepwise regression: *RestingBP*, *HeartDisease*, *FastingBS*, *Cholesterol*, and *Oldpeak*, in that order. The final model retained *Age*, *Sex*, *ChestPainType*, *RestingECG*, *ExerciseAngina*, and *ST_Slope* as predictors ($R^2 = 0.328$, Adjusted $R^2 = 0.319$), explaining approximately 32.8% of the variance in maximum heart rate. A one year increase in age is generally associated with a decrease of approximately 0.77 bpm in maximum heart rate ($\beta = -0.766$, $p < 0.001$). Male patients have lower maximum heart rates than female patients ($\beta = -3.57$, $p = 0.045$). All chest pain types show significantly higher maximum heart rates compared to the asymptomatic reference group, and patients with exercise-induced angina have significantly lower maximum heart rates ($\beta = -6.80$, $p < 0.001$). Though its individual level p -values were non-significant, *ST_Slope* was retained based on a significant ANOVA F -test ($p < 0.001$).

Residual Plots





The fits vs. residuals plot shows residuals randomly scattered around zero with no obvious pattern or megaphone effect, suggesting the linearity and constant variance assumptions are reasonably satisfied. Only 3 points have studentized residuals exceeding 3, indicating very few extreme outliers. The Normal Quantile plot shows the residuals follow the normal line closely through the middle of the distribution, with minor deviations at both ends indicating slight departures from normality. Overall we can consider the model assumptions to be reasonably met.



MISC

Prior to modeling, the distribution of serum cholesterol was noted to be right-skewed, suggesting a log transformation might be appropriate. However, cholesterol was removed during backwards stepwise regression due to its lack of significance as a predictor of $MaxHR$, so no transformation was ultimately necessary. A Box-Cox analysis of the final model indicated that $\lambda = 1$ (no transformation) fell outside the 95% confidence interval, suggesting a power transformation of $MaxHR$ with $\lambda \approx 1.5$ may be appropriate. A transformed model was fit using $MaxHR^{1.5}$ as the response variable, which improved normality of the residuals. However, the transformation introduced slight heteroskedasticity in the fits vs. residuals plot that was not present in the original model. Given that heteroskedasticity is a more consequential violation of regression assumptions than mild non-normality, and that the large sample size ($n = 746$) provides robustness to minor departures from normality, and that $MaxHR$ is considerably easier to interpret than $MaxHR^{1.5}$ the untransformed model was retained as the final model.

Analysis - Logistic Regression

Starting from the full model, predictors were removed in the following order:
RestingECG ($p = 0.631$), *FastingBS* ($p = 0.363$), *Cholesterol* ($p = 0.163$), and
RestingBP ($p = 0.058$). The final model summary is shown below.

Final Model

```
##
## Call:
## glm(formula = HeartDisease ~ Age + Sex + ChestPainType +
##      ExerciseAngina +
##      Oldpeak + ST_Slope, family = binomial, data = heart_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.60085    0.94673  -3.803 0.000143 ***
## Age             0.03891    0.01316   2.958 0.003098 **
## SexM           1.79156    0.29957   5.980 2.22e-09 ***
## ChestPainTypeATA -1.65550    0.34450  -4.806 1.54e-06 ***
## ChestPainTypeNAP -1.57932    0.29584  -5.338 9.38e-08 ***
## ChestPainTypeTA  -1.51087    0.46339  -3.260 0.001112 **
## ExerciseAnginaY  0.93541    0.26059   3.590 0.000331 ***
## Oldpeak        0.42148    0.13782   3.058 0.002227 **
## ST_SlopeFlat    1.23923    0.51047   2.428 0.015198 *
## ST_SlopeUp      -1.27465    0.55154  -2.311 0.020828 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1032.63  on 745  degrees of freedom
## Residual deviance:  490.74  on 736  degrees of freedom
## AIC: 510.74
##
## Number of Fisher Scoring iterations: 5

##              OR      2.5 %   97.5 %      p
## (Intercept)  0.0273006 0.0041511  0.1712 0.0001427 ***
## Age          1.0396820 1.0134343  1.0672 0.0030977 **
## SexM         5.9987779 3.3745424 10.9430 2.225e-09 ***
```

```
## ChestPainTypeATA 0.1909974 0.0955195 0.3702 1.543e-06 ***
## ChestPainTypeNAP 0.2061147 0.1142695 0.3653 9.375e-08 ***
## ChestPainTypeTA 0.2207188 0.0875071 0.5413 0.0011123 **
## ExerciseAnginaY 2.5482582 1.5272540 4.2503 0.0003312 ***
## Oldpeak 1.5242186 1.1694146 2.0095 0.0022272 **
## ST_SlopeFlat 3.4529652 1.2518038 9.3270 0.0151976 *
## ST_SlopeUp 0.2795279 0.0934472 0.8170 0.0208278 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: HeartDisease
##
## Terms added sequentially (first to last)
##
##
##              Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
## NULL              745      1032.63
## Age                1    69.364      744    963.26 < 2.2e-16 ***
## Sex                1    67.701      743    895.56 < 2.2e-16 ***
## ChestPainType      3   175.571      740    719.99 < 2.2e-16 ***
## ExerciseAngina     1    79.407      739    640.58 < 2.2e-16 ***
## Oldpeak            1    48.162      738    592.42 3.923e-12 ***
## ST_Slope           2   101.676      736    490.74 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Results & Discussion

A binary logistic regression model was applied to predict the presence of heart disease among patients. We then applied manual backward stepwise regression to remove RestingECG, FastingBS, Cholesterol, RestingBP in that order, and obtained Age, Sex, ChestPainType, ExerciseAngina, Oldpeak, and ST_Slope as significant predictors. AIC further supports the model selection, since the final model (AIC=510.74) has a lower value than the full model (AIC=513.59), confirming that the removed predictors weren't contributing meaningfully to model fit. All predictors in the final model are significant at the $\alpha = 0.05$ level, and ST_Slope was retained based on a significant

deviance test ($p < 0.001$) despite its individual level p -values being only marginally significant.

The deviance table confirms that each predictor in the final model contributes significantly to the model (all $p < 0.001$). The null deviance was 1032.63, which dropped to 490.74 when all significant predictors were included. This reduction of 541.89 shows that the model explains substantially more variation in heart disease status than the null model.

For odds ratios, men have approximately 6 times higher odds than women to have heart disease ($OR = 6.00$, $p < 0.001$). For every one year increase in age, the odds of heart disease increase by 4% ($OR = 1.04$, $p = 0.003$). People who experience angina when exercising are about 2.5 times more likely to have heart disease ($OR = 2.55$, $p < 0.001$). For each unit increase in $oldpeak$ the odds of heart disease increase by 1.52 ($OR = 1.52$, $p = 0.002$). All types of chest pain have significantly lower odds of heart disease than the asymptomatic reference group (OR Range = 0.19 - 0.22). Interestingly, this counter intuitive result is similar to a well known clinical observation: asymptomatic patients tend to have worse underlying heart disease, precisely because they have no warning symptoms. A flat ST slope is associated with 3.45 times the odds of heart disease ($OR = 3.45$, $p = 0.015$) when compared to the down-sloping reference level, whereas an up-sloping ST slope is associated with lower odds ($OR = 0.28$, $p = 0.021$).

To demonstrate the model's predictive power, consider patient 9, a 37-year-old man with asymptomatic chest pain, angina on exertion, and flat ST slope. The model predicts there is a 92% chance he has heart disease, which is consistent with his actual diagnosis.

Conclusions

In this analysis, clinical predictors of heart disease were examined using data from 746 patients. Descriptive analysis suggested that heart disease patients have generally lower maximum heart rates across all ages. Multiple regression showed that age, sex, type of chest pain, ECG results, exercise induced angina, and ST slope were significant predictors of maximum heart rate ($R^2 = 0.328$). Binary logistic regression analysis showed similar predictors of heart disease presence, with male sex, asymptomatic chest pain, exercise induced angina, and flat ST slope being associated with significantly increased odds of heart disease. Collectively, these results indicate that clinical measures related to exercise, especially maximum heart rate, ST slope, and exercise-induced angina, are among the most informative markers for heart disease in this data set.